

JEE MAIN REVISION MASTER: D AND F BLOCK ELEMENTS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	D And F Block Elements — Core Principles & Governing Laws	22 PYQs	Core Principle
Moderate	D And F Block Elements — Applied Cases, Reactions & Traps	5 PYQs	Applied Topics
Standard	D And F Block Elements — Standard Calculations & Problem Archetypes	2 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
D And F Block Elements — Core Principles & Governing Laws	$\text{Cu}_{2}\text{O}, \mu=\sqrt{n(n+2)}\text{B.M}, \text{Cr}^{2+}$
D And F Block Elements — Applied Cases, Reactions & Traps	$\text{CrO}_{2}, \text{Mn}^{2+}, \text{K}_{2}\text{Cr}_{2}\text{O}_{7}$
D And F Block Elements — Standard Calculations & Problem Archetypes	$[\text{Ar}]3d^{10}4s^{1}, (n-1)d^{10}ns^{2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 D And F Block Elements — Core Principles & Governing Laws (Very High Yield — 22 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in D And F Block Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in D And F Block Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in D And F Block Elements PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The calculated spin only magnetic moment of Cr^{2+} ion is..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "The highest oxidation state of manganese in fluoride is +4 (MnF_4), but the highest oxidation state in oxides is +7 (Mn_2O_7)" [Ans: D]
- [2026 Shift] Q#3 (MCQ): "Which of the following will not act as an oxidising agent?" [Ans: N/A]

3.2 D And F Block Elements — Applied Cases, Reactions & Traps (Moderate Yield — 5 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of D And F Block Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "Which of the following statements related to lanthanoids is incorrect?" [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): "Which of the following does not represent property stated against it?" [Ans: N/A]
- [2019 Shift] Q#1 (MCQ): "Incorrect statement with reference to Ce ($Z=58$) is..." [Ans: N/A]

3.3 D And F Block Elements — Standard Calculations & Problem Archetypes (Standard Yield — 2 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in D And F Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "In which of the following pairs, both the elements do not have $(n-1)d^1ns^2$ configuration?" [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "What will be the value of x in Fe^{x+} , if the magnetic moment, $\mu = \sqrt{24} \text{ BM}$?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The calculated spin only magnetic moment of Cr^{2+} ion is

- (A) 3.87 BM
- (B) 4.90 BM **[Correct]**
- (C) 5.92 BM
- (D) 2.84 BM

Correct Answer: B

Solution Analysis: For spin only magnetic moment, we use the formula $\mu = \sqrt{n(n+2)}\text{BM}$ where n is the number of unpaired electrons. Chromium has atomic number 24. Its electronic configuration is $\text{Cr} = [\text{Ar}]3d^5 4s^1$. For Cr^{2+} , two electrons are removed. First the 4s electron is removed, then one electron from 3d. So, $\text{Cr}^{2+} = [\text{Ar}]3d^4$. In $3d^4$, according to Hund's rule, the four electrons occupy four different d orbitals singly. So the number of unpaired electrons is $n=4$. Now, $\mu = \sqrt{4(4+2)} = \sqrt{24}$ $\mu = 4.90\text{BM}$. So the correct answer is B.

Benchmark Problem 2: KCET 2025 (Q#2)

In which of the following pairs, both the elements do not have $(n-1)d^1 0ns^2$ configuration?

- (A) Cu, Zn
- (B) Zn, Cd
- (C) Cd, Hg
- (D) Ag, Cu

Correct Answer: None

Solution Analysis: To determine which pairs of elements do not both have the configuration $(n-1)d^1 0ns^2$, let's examine the electronic configurations provided: Cu (Copper): $[\text{Ar}]3d^9 4s^1$, Zn (Zinc): $[\text{Ar}]3d^{10} 4s^2$, Cd (Cadmium): $[\text{Kr}]4d^{10} 5s^2$, Hg (Mercury): $[\text{Xe}]4f^{14} 5d^{10} 6s^2$, Ag (Silver): $[\text{Kr}]4d^{10} 5s^1$. From the configurations: Copper (Cu) does not have an ns^2 configuration (as it is $4s^1$). Silver (Ag) also lacks an ns^2 configuration (as it is $5s^1$). Thus, in the pair consisting of Ag and Cu, neither element has the $(n-1)d^1 0ns^2$ configuration.

Benchmark Problem 3: KCET 2024 (Q#2)

Which of the following statements related to lanthanoids is incorrect?

- (A) Lanthanoids are silvery white soft metals.
- (B) Samarium shows +2 oxidation state.
- (C) Ce^{+4} solutions are widely used as oxidising agents in titrimetric analysis.
- (D) Colour of lanthanoid ion in solution is due to d-d transition.

Correct Answer: None

Solution Analysis: The incorrect statement regarding lanthanoids is given in option (d). Its correct form is colour of Lanthanoid ion in solution is due to f-f transition.